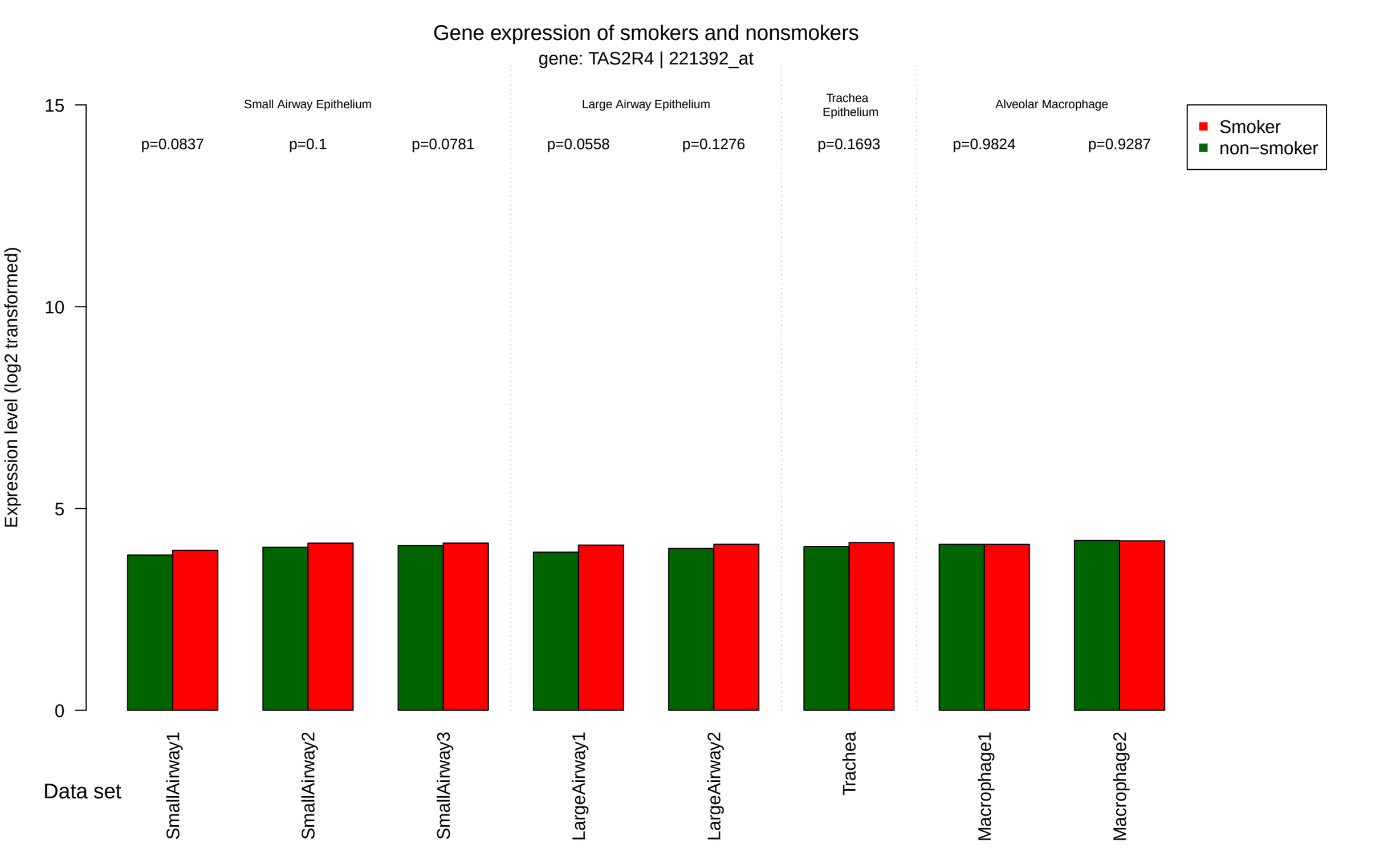
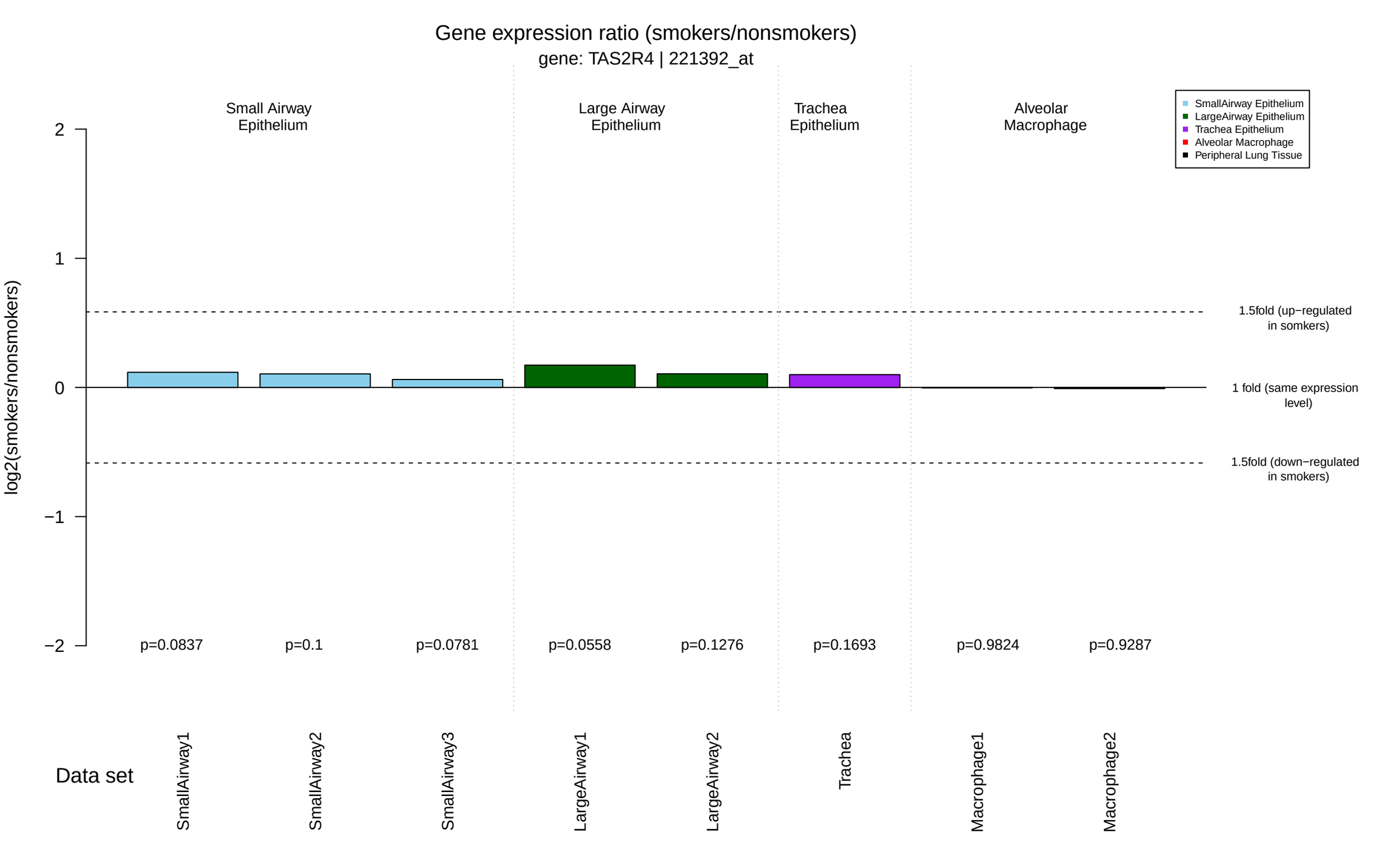
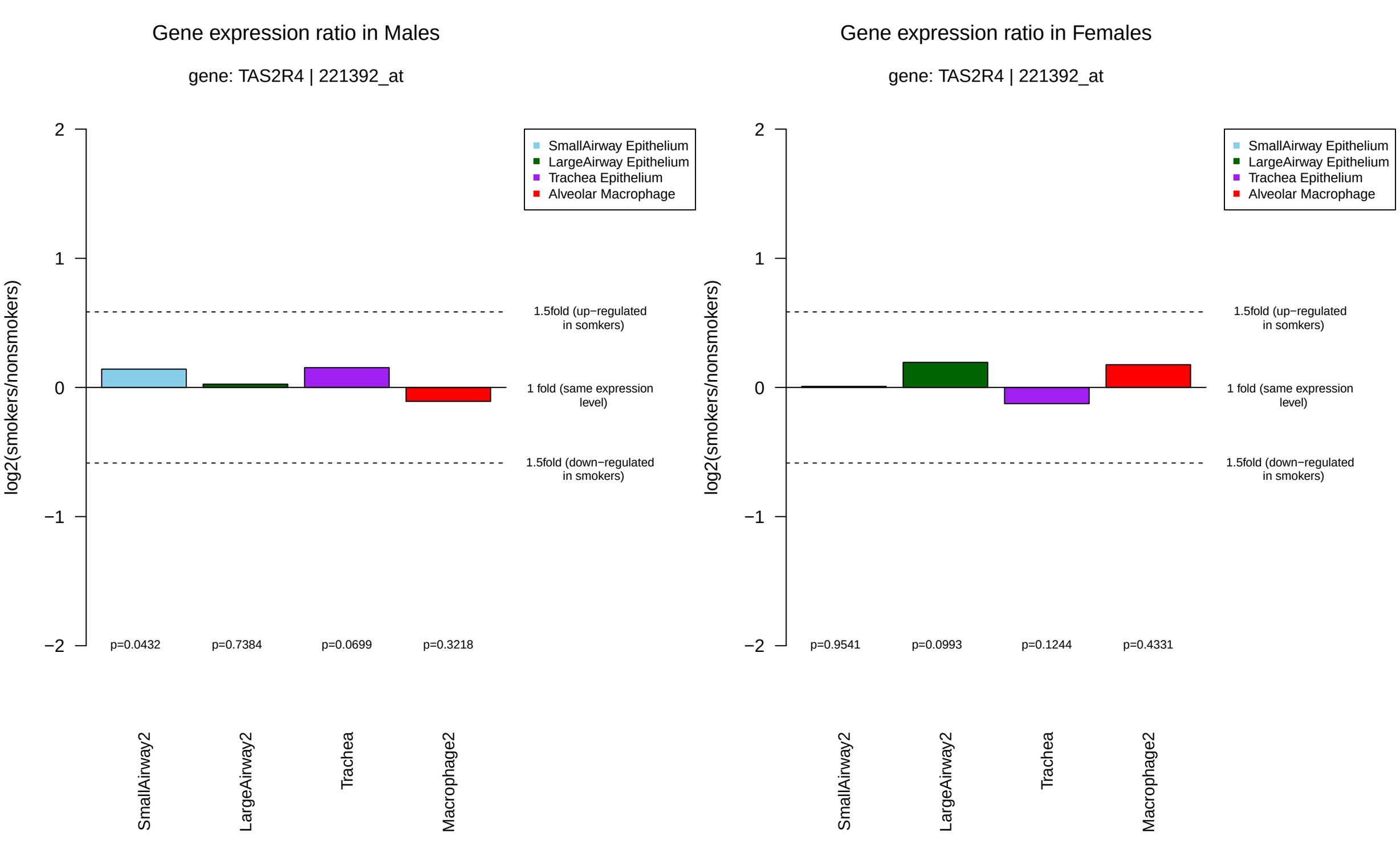
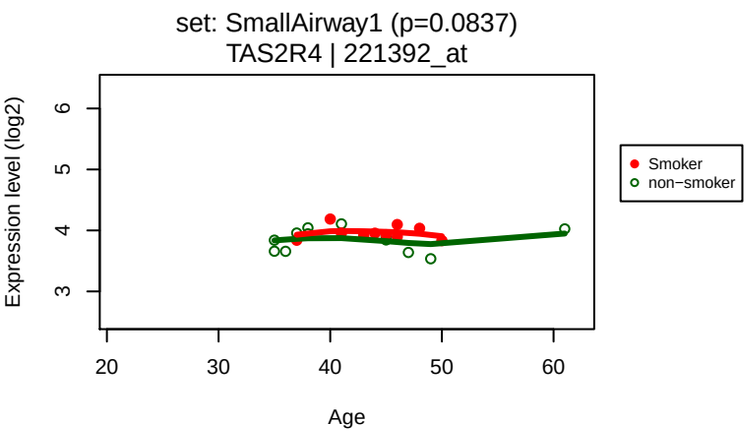
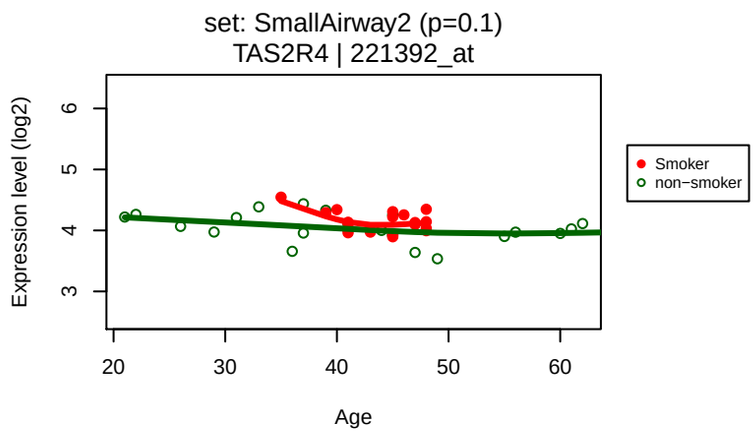
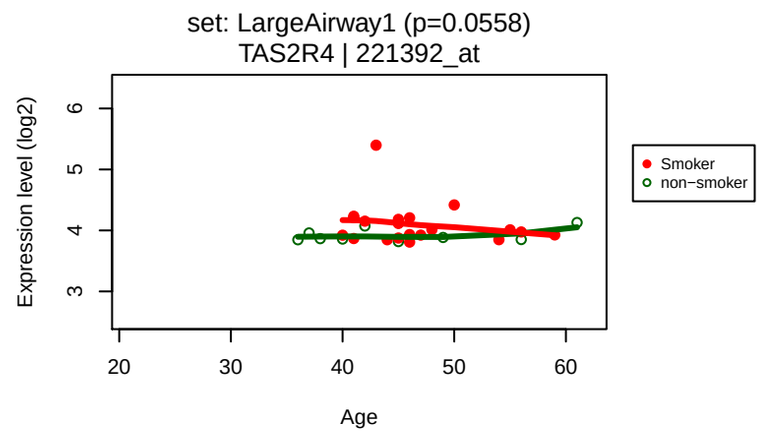

Gene expression of smokers and nonsmokers
gene: TAS2R4 | 221392_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0837
p=0.1
p=0.0781
p=0.0558
p=0.1276
p=0.1693
p=0.9824
p=0.9287
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

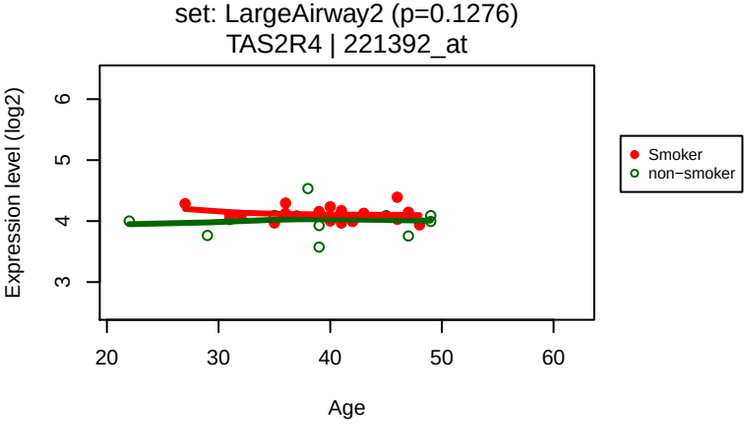

Gene expression ratio (smokers/nonsmokers)
gene: TAS2R4 | 221392_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0837
p=0.1
p=0.0781
p=0.0558
p=0.1276
p=0.1693
p=0.9824
p=0.9287
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

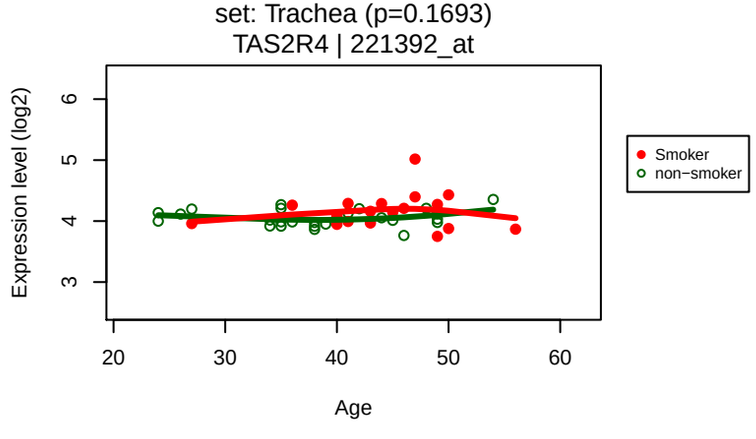

Gene expression ratio in Males
gene: TAS2R4 | 221392_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0432
p=0.7384
p=0.0699
p=0.3218
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAS2R4 | 221392_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9541
p=0.0993
p=0.1244
p=0.4331
SmallAirway2
LargeAirway2
Trachea
Macrophage2

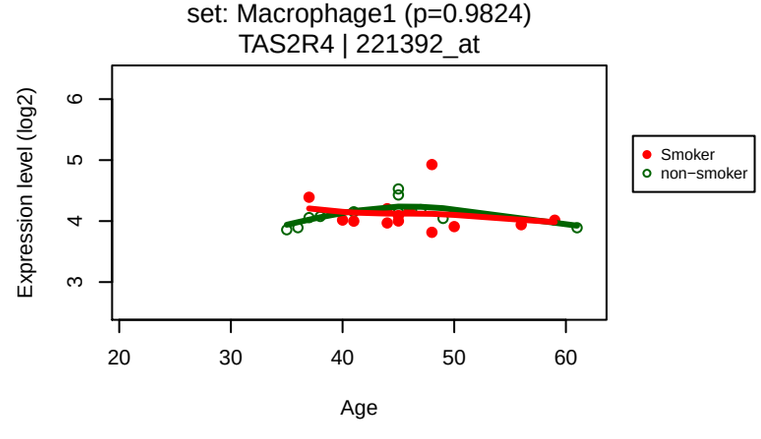

set: SmallAirway1 (p=0.0837)
TAS2R4 | 221392_at
Expression level (log2)
Age
Smoker
non−smoker

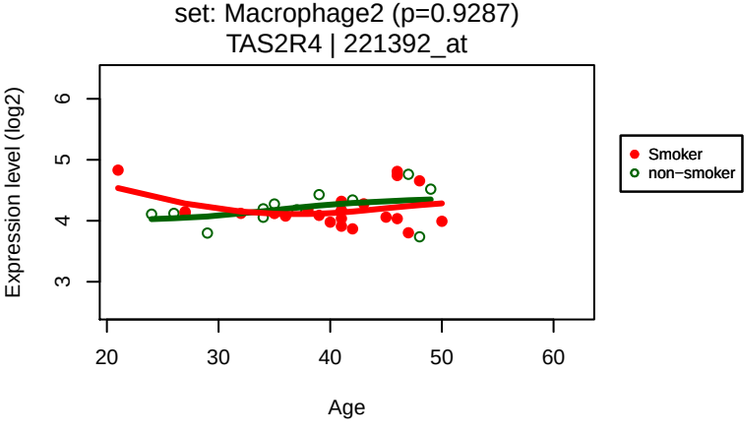

set: SmallAirway2 (p=0.1)
TAS2R4 | 221392_at
Expression level (log2)
Age
Smoker
non−smoker

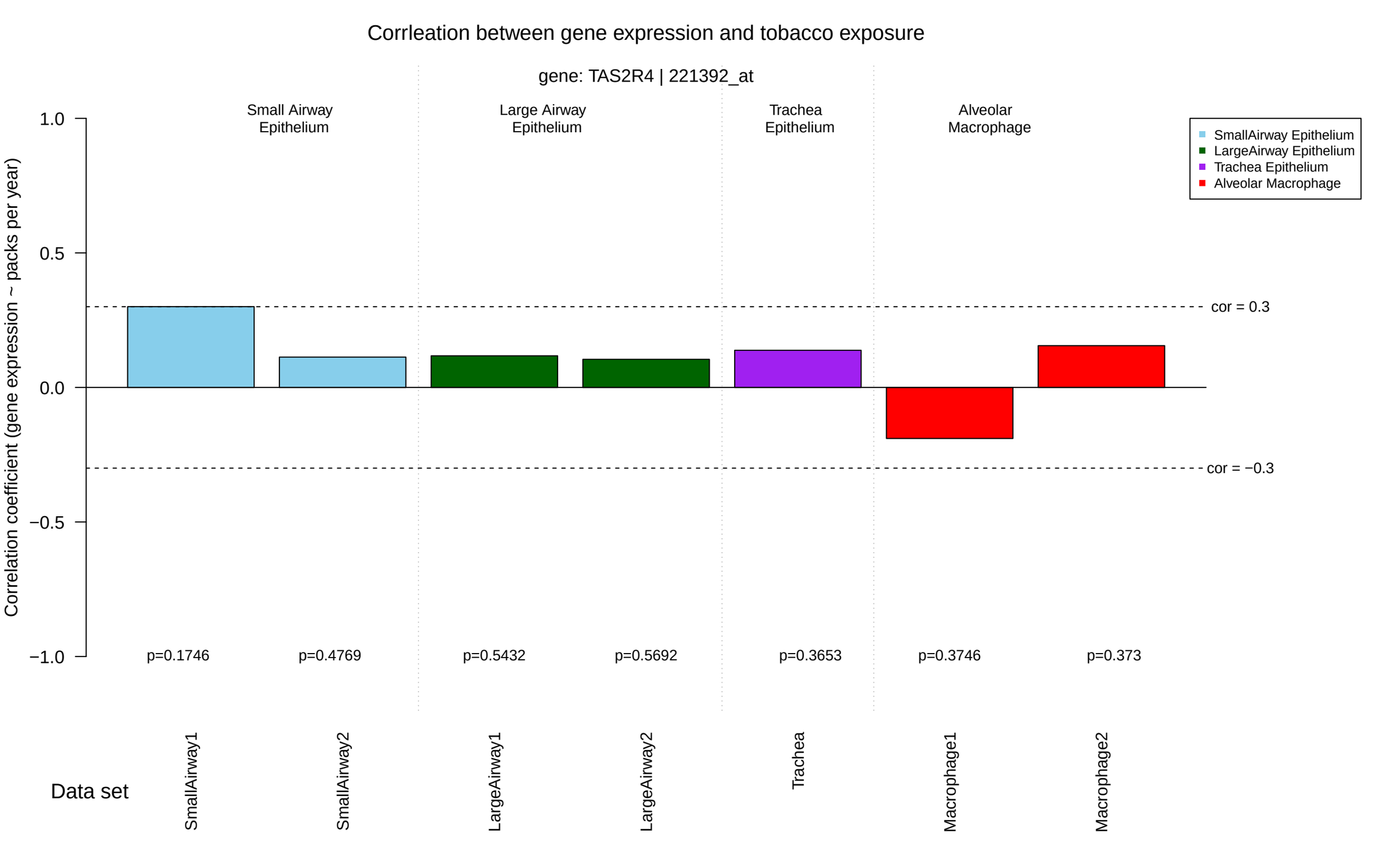

set: LargeAirway1 (p=0.0558)
TAS2R4 | 221392_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1276)
TAS2R4 | 221392_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1693)
TAS2R4 | 221392_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9824)
TAS2R4 | 221392_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9287)
TAS2R4 | 221392_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAS2R4 | 221392_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1746
p=0.4769
p=0.5432
p=0.5692
p=0.3653
p=0.3746
p=0.373
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2